Appendix

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A Psychological Hooks Operationalization

A.1 FOMO

Przybylski et al. [?] described "fear of missing out" as a prevalent concern, specifically the worry that others might be experiencing beneficial events that one is not part of, characterized by a desire to continuously know what others are doing. In McGuire's motivational matrix [?], the tension reduction theory (dimension: stability-active-emotional state-internal relationship) depicts humans as an energy system that derives pleasure from the release of tension and feels pain from the increase of tension. Additionally, his utility theory (dimension: growth-reactive-cognitive state-internal relationship) emphasizes people's need to pursue maximum benefits at the lowest cost.

Formally, content information cards that include the fear of missing out typically highlight the potential losses of missing out, arousing the audience's worries and increasing their tension. This thereby creates motivations such as tension reduction and cost minimization, which can be satisfied through clicking and further reading.

Core Definition: The content expresses "if you don't do something, there will be certain consequences (losses / regrets)" to arouse the audience's concerns and increase tension.

Operational Judgment: Does the content contain/convey: clues of "inaction" AND "costs" or "consequences" of "inaction"?

- **Clues of "inaction":** Words like not watching / not listening / not doing / not acting.
- **Costs and consequences:** Negative intentions such as bankruptcy, breakup, or failure.

A.2 Gain Appeal

As mentioned above, the utility theory in McGuire's motivational matrix describes how people derive satisfaction by maximizing benefits at the lowest cost. Meanwhile, his assertion theory (dimensions: growth-active-emotional state-external relationship) [?] also emphasizes people's needs to enhance their abilities and pursue improvement.

Formally, content information cards that include resource acquisition usually highlight the value that their information can bring, stimulating the audience's inherent motivation to acquire, thereby prompting clicks.

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coredefThe content emphasizes "what benefits can be obtained through this content information" to stimulate the audience's inherent motivation to acquire.

Operational Judgment: Does the content contain/convey: the benefits that the content can bring?

- **Benefits:** Money (saving/making money), time (improving efficiency), health (weight loss/beauty), skills (quick learning), and emotions (happiness/peace).

A.3 Information-gap

Loewenstein [?] proposed a new explanation of curiosity, interpreting it as a cognitively induced sense of deprivation that arises from people's perception of gaps in their knowledge or understanding. He also pointed out that Gestalt psychologists have long been among the strongest advocates of the view that "humans have a need to understand". In fact, the concept of "Gestalt" itself reflects the basic tendency of humans to understand information by organizing it into a coherent "whole". Finally, he summarized five situational factors that may arouse this type of curiosity, including: posing questions (i.e., Berlyne's "thematic probes" [?]), predictable but unknown information, information inconsistency, information gaps between people, and information gaps between the past and the present. These provide a direct reference for our operational definition.

The stimulus theory in McGuire's motivational matrix (dimension: growth-active-cognitive state-internal relationship) [?] also describes people's need for exploration to a certain extent.

Formally, content information cards containing information gaps can take many forms, which can be directly reflected in the title or the cover. The most common form is "thematic probes": posing questions and riddles to arouse the audience's curiosity and desire to fill in the gaps.

Core Definition: When the content could have fully conveyed information through the title or cover, it intentionally omits part of the information to guide the audience to click.

Operational Judgment: Does the content create: intentionally hidden information?

- **Self-questioning and answering:** Raising a question in the title and answering it in the content.
- **Obscuring information:** Using mosaics or stickers to cover key parts.
- **Ambiguous reference:** Using words like "this" or "that" without clear referents.

A.4 Anomaly and novelty

Berlyne [?] pointed out that the type of curiosity that enhances the perception of stimuli and the type of curiosity whose main outcome is knowledge are likely to prove to be closely related, namely the perceptual curiosity and cognitive curiosity he classified. Perceptual curiosity refers to a drive triggered by novel stimuli and diminished with continued exposure to these stimuli, which is consistent with the stimulus theory in McGuire's motivational matrix [?], emphasizing that people are curious novelty seekers

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and eager to avoid boredom. Cognitive curiosity refers to the desire for knowledge and is mainly applicable to humans.

He believed that when studying cases of strange, unusual, and confusing things, it is necessary to use the variable of conflict for explanation, and attributed the curiosity aroused by such situations or problems to learned conflict. Berlyne [?] also discussed some integrative stimulus characteristics, i.e., those stimuli that can arouse or induce curiosity drive. A stimulus can be in different positions on dimensions such as familiar-novel, expected-unexpected, simple-complex, clear-ambiguous, and other dimensions.

The consistency theory (dimensions: stability-active-cognitive state-internal relations) in McGuire's motivational matrix also emphasizes that people strive to maintain an interconnected and coherent cognitive system. When there is any imbalance between one's own perceptions, memories, emotions, needs, behaviors, role commitments, cultural norms, etc., people will be driven to take action to reduce this imbalance. At the same time, the process of this action is also a process of seeking explanations and generating meanings, which is also driven by the motivation of hermeneutic theory (dimensions: stability-active-cognitive state-external relations).

As Berlyne said, "associative variables" such as "novelty" and "complexity" depend on the comparison of stimulus characteristics, the organization of information from different sources, and the examination of similarities and differences. Therefore, here we operationally define "abnormality" and "novelty" in the psychological hook of abnormal novelty as whether the content author "packages" the content as abnormal or novel, rather than judging whether the content is really abnormal or novel for a certain audience.

Therefore, formally, information cards containing abnormal novelty will directly claim or describe the content as surprising, beyond imagination, or rare and novel through some form of packaging.
Core Definition: Content is deliberately packaged to be astonishing, counterintuitive, rare, or novel to arouse curiosity.
Operational Judgment: Does the content contain: phrases expressing astonishing abnormality OR rarity and novelty?

- **Abnormality:** Phrases like "unexpectedly", "incredible", or "refreshing one's outlook".
- **Rarity/Novelty:** Extreme words (most, top, epic-level) or scarcity words (only, limited, niche).

A.5 Perceptual Contrast

Rosch [?] verified a hypothesis through experiments, which is: natural categories (such as colors, line directions, and numbers) have reference point stimuli (such as focal colors, vertical and horizontal lines, and numbers that are multiples of 10), and other stimuli in the category are judged with reference to these reference point stimuli. Earlier, other scholar proposed that in perceptual stimuli, there exist certain "ideal types" that act as anchors for perception. Therefore, when two or more completely different things or states are juxtaposed, they serve as a stimulus. This stimulus not only arouses people's need for classification (dimension: stability-reaction-cognitive state-internal relationship), that is, people will classify the received impressions into their already formed cognitive categories in complex situations, but also causes conflict and cognitive dissonance, stimulating the need for consistency and the need for explanation in McGuire's motivational matrix.

Formally, information cards containing perceptual contrast will use some kind of contrast (this contrast is inherently anchored) through image contrast, semantic contrast, or a mixed method (i.e., contrast between images and semantics) to arouse the audience's curiosity to "find out more".
Core Definition: Content places two or more contrasting states or things together visually or textually to stimulate desire to explore.
Operational Judgment: Does the content contain/convey: two or more contrasting items with obvious differences?

- **Contrasting items:** Front and back, positive and negative, expectation vs. reality.
- **Forms:** Semantic contrast in text, visual contrast in images, or text-image contrast.

A.6 Ingroup Identification / Outgroup Distinction

Intergroup categorization networks are ubiquitous in social environments; they permeate our socialization and educational processes, from "teams" and "team spirit" in primary and secondary schools, to various adolescent groups, and further to social, national, ethnic, racial, religious, or age groups [?]. The identity theory (dimension: growth-reactive-emotional state-internal relationship) in McGuire's motivational matrix integrates perspectives from multiple schools of thought, emphasizing that individuals can elaborate on their self-awareness and gain satisfaction through social roles and group belonging.

In the process of group belonging and identification, it may also be influenced by how people divide groups based on the template theory in their own cognition, how people make inductive judgments through certain behaviors, and how people are "infected" and synchronized by others. These points correspond respectively to the template theory (dimension: growth-reactive-cognitive state-external relationship), induction theory (dimension: stability-reactive-cognitive state-external relationship), and infection theory (dimension: growth-reactive-emotional state-external relationship) in McGuire's motivational matrix.

In addition, Tajfel H et al. [?] also described the competition and conflict between ingroups and outgroups, pointing out that these two forms of intergroup conflict are in a complex interdependent relationship and reinforce each other.

Therefore, formally speaking, information cards containing group identity will first display group labels in some way, thereby arousing ingroup identity or creating teasing or exclusion of outgroups.
Core Definition: Content uses group labels to arouse a sense of identity and belonging; or to arouse rejection and ridicule towards a certain group.
Operational Judgment: Does it contain group labels AND (display attitudes of belonging/rejection OR calls to action OR group commonalities)?

- **Group Labels:** Nouns related to race, country, religion, age, gender, occupation, or MBTI tags.
- **Attitudes/Actions:** Showing pride or empathy; disdain or satire; calls like "like if you are...".

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A.7 Social Comparison

Blanton et al. [?] pointed out that social comparison has multiple functions, and people will change their comparison strategies according to their current motivations. Specifically, when people face threats to their self-esteem that cannot be resolved through instrumental actions, they will seek to compare themselves with those who are in a worse situation; when they face threats that they can cope with and the desire for self-improvement prevails, they will seek upward comparisons. The assertion theory in McGuire's motivational matrix also emphasizes people's competitive instinct and the need to gain a sense of superiority and dominance; his self-defense theory (dimension: stability-reaction-emotional state-internal relationship) emphasizes the necessity for people to maintain self-esteem.

Gibbons F.X. and Buunk B.P. [?] pointed out that although social comparison is automatic, meaning that almost everyone engages in social comparison from time to time, it is also situational. For most people, situations that foster competition may arouse their interest in social comparison, and uncertain or threatening situations will also increase individuals' tendency to engage in social comparison.

Formally, information cards containing social comparison will create situations that stimulate social comparison (such as showing differences in social resources, directly using comparison-related words, etc.) or trigger the audience to participate in comparison by displaying attitudes after upward/downward comparisons.

Core Definition: Content triggers the audience to engage in comparison by using comparative words, showing gaps and displaying certain attitudes.

Operational Judgment: Are there obvious comparative action words OR (display of gaps AND display of attitudes)?

- **Action words:** "than...", "VS", "not as good as...", "crush" .
- **Gaps:** Ability gaps (achievements), self-trait gaps (appearance), or resource gaps (wealth).
- **Attitudes:** Upward (jealousy vs. motivation) or Downward (showing off vs. contentment).

A.8 Authority Endorsement

According to the template theory in McGuire's motivational matrix, the authoritative role template provides a paradigm for behaviors or beliefs. Meanwhile, the assertion theory emphasizes people's desire for self-improvement. "Gladiators" will admire and imitate the strong, and this imitation, to a certain extent, aligns with the satisfaction gained through social learning as emphasized by the contagion theory.

Therefore, formally speaking, information cards containing authoritative endorsement will use various powerful source endorsements to exempt the audience from the cost of questioning, thereby making the audience convinced and even imitating.

Core Definition: Content uses various persuasive source endorsements to exempt the audience from questioning costs, guiding them to be convinced.

Operational Judgment: Does the content contain: source endorsement?

- **Authoritative sources:** Experts, institutions, celebrities, rankings, or certifications (e.g., FDA, Harvard research).
- **Social proof:** Numbers such as "220,000 people have viewed...".

B Invalid Output Analysis

Table 1: invalid output rate across different models

Model	Invalid Output Rate (%)
Qwen2.5-VL-3B-Instruct	4.20
Qwen2.5-VL-7B-Instruct	5.47
Qwen2.5-VL-32B-Instruct	4.72
DeepSeek-VL2	1.15
GLM-4.1V-9B-Thinking	1.08
Gemma3-4b	0.31
Gemini 2.0 Flash	2.36
Claude 3.7 Sonnet	1.32
Yi-Vision-v2	1.98

As a supplementary metric for model reliability, Table 1 reports the Invalid Output Rate for each evaluated model. This metric represents the proportion of model responses that failed to conform to the required output format (e.g., garbled text or incorrect JSON structure).

C Performance Different Annotators

To ensure the reliability of the dataset and filter out potential malicious annotations, we calculated the mean, variance, and standard deviation of the scores assigned by five individual human annotators. This statistical oversight ensures that each annotator's output remains within a reasonable and consistent range.You can see the result in Table 2.

D Annotation Platform Interface

Our specialized annotation platform provides a structured environment for the annotators. The interface The interface, as illustrated in Figure 2, displays comprehensive image-text content, operational definitions, scoring criteria, and intuitive selection boxes to facilitate precise labeling.

E Model Performance Across Vertical Categories

To evaluate the model's robustness, we calculated the F1-score of the test results across different vertical data samples. The detailed performance distribution across various categories (such as career, cosmetics, and food) can be seen in the heatmap provided in Figure 1.

F Statistical Characterization of Psychological Hooks

In this section, we provide the detailed statistical distribution of the identified psychological hooks within our dataset. Table 3 summarizes the mean (μ), variance (σ^2), and standard deviation (σ) for each of the eight predefined hook categories. These indicators characterize the frequency and dispersion of specific hooks across the sampled content, providing a quantitative basis for the complexity analysis discussed in the main text.

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G Dataset Composition and Construction Details

Table 4 provides a detailed breakdown of the number of psychological hooks identified per sample, distinguishing between single-hook, composite-hook (including the specific distribution of multiple labels), and no-hook instances. This granular view supplements the general dataset statistics by highlighting the co-occurrence frequency of different psychological mechanisms.Table 5 illustrates the distribution of data across various social media content categories. Table 6 provides a multifaceted overview of the dataset scale, including the distribution of image and video modalities, alongside linguistic and engagement metrics such as title lengths and like counts. To further characterize the psychological landscape of our data, Table 7 enumerates the occurrences of the eight predefined hook categories and their respective annotation quality levels, distinguishing between high-consensus samples and complex edge cases. Together, these statistics offer a granular foundation for understanding the data diversity and labeling rigor underlying our research.

The Dataset Construction Pipeline. Figure 3 illustrates the systematic workflow employed to build the PsyHookBench dataset, covering stages from raw data collection to final expert verification. It offers a visual roadmap of our methodology to ensure the reproducibility and quality of the annotation process.

H Detailed Related Work Supplement

With the increasing complexity of model inputs, several benchmarks have been developed to evaluate interleaved multimodal sequences and temporal integration. For instance, **Seed-Bench** [?] emphasizes video understanding and temporal logic, while **BLINK** [?] introduces visual prompts to test sensitivity to critical visual evidence. Additionally, **NLVR2** [?] is frequently employed to assess fine-grained discrimination through paired-image reasoning. These benchmarks collectively establish a multi-dimensional landscape for evaluating the technical limits of VLMs across various input modalities and task complexities.

Theory of Mind (ToM), the ability to understand one's own and others' beliefs, desires, and intentions, is considered central to human social interaction and empathy. Research by Kosinski [?] indicates that ToM-like abilities may emerge spontaneously as a byproduct of the improvement in language model capabilities. Their study found that GPT-3.5 (davinci-003) can perform at the level of a 9-year-old child in false belief tasks. Bubeck et al. [?] also discovered in their early experiments on GPT-4 that the model possesses the ability to understand others' beliefs, desires, and intentions.

Hagendorff et al. [?] systematically studied the cognitive biases of Large Language Models (LLMs) and found that as the model scale increases, the models exhibit cognitive errors similar to those in human intuitive system thinking; however, newer models such as ChatGPT can correct intuitive biases through reasoning abilities demonstrated by Chain-of-Thought (CoT). In the field of visual cognition, Zellers et al. [?] noted that models still face challenges in higher-level cognitive tasks in visual understanding, such as inferring the behavioral intentions and mental states of people in images.

Furthermore, Coda-Forno et al. [?] introduced methods from computational psychiatry and found that GPT-3.5 not only responds strongly to anxiety questionnaires but also, after being prompted to enter an anxious state, shows impaired exploratory decision-making ability and stronger racial and ability biases. This finding confirms the significant impact of prompt engineering on the model's mental state and subsequent behaviors. To more comprehensively assess the human-like characteristics of LLMs, Huang et al. [?] proposed an integrated PsychoBench framework, which includes 13 clinical psychological scales covering four dimensions: personality traits, interpersonal relationships, motivation, and emotional abilities. This study not only evaluated mainstream models but also explored the issue of validating assessment scales on LLMs through role-playing, proving the correlation between scale scores and actual behaviors.

I Definitions of Ethical Risks Detail

Following the taxonomy of Biyani et al. (2016) and the specific ecological characteristics of Xiaohongshu, we define the four categories of ethical risks as follows:

- **Omission**: Content where the headline promises specific information (e.g., "The secret to...") that is completely absent from the actual post or images.
- **Mismatch**: The content promised in the title and cover contradicts or fails to align with the actual expanded content.
- **Inflammatory**: The use of vulgar language, excessive emotional manipulation, or provocative phrasing designed to incite conflict or verbal abuse.
- **Adverse Influence**: Content promoting illegal activities, harmful social trends, or behavior that violates established moral standards and platform community guidelines.

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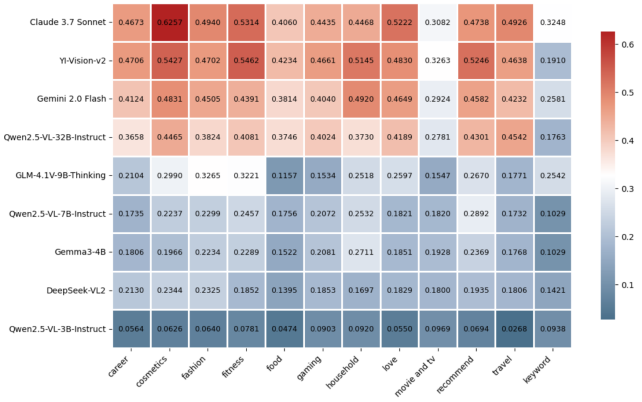

Figure 1: Performance heatmap of different models across various categories

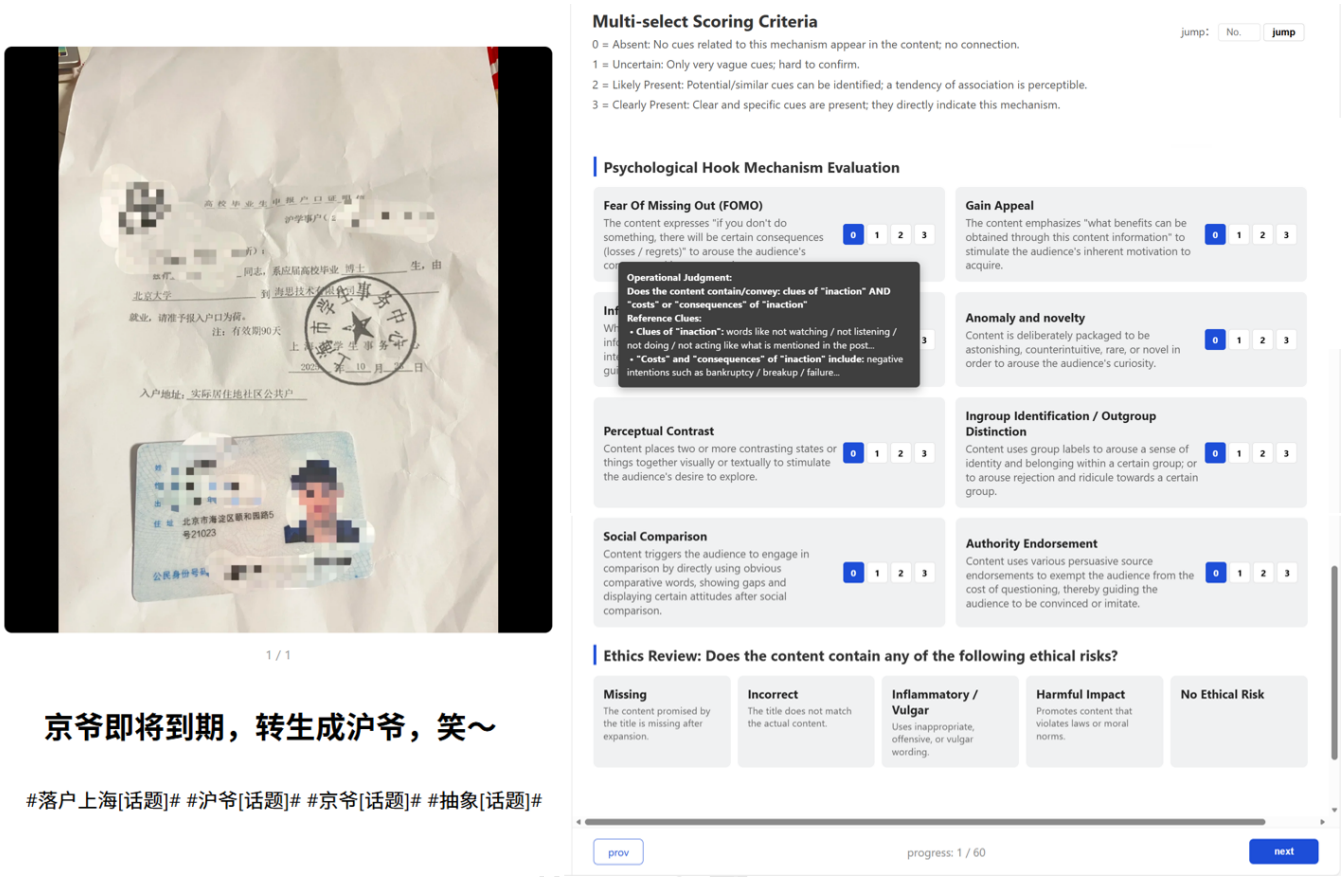

Figure 2: Annotation platform GUI

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Table 2: Performance statistics for different annotators

Statistic	Ann. 1	Ann. 2	Ann. 3	Ann. 4	Ann. 5
Mean	0.8571	0.2710	0.4368	0.5816	0.4354
Variance	1.3946	0.6114	0.8050	1.2292	0.8870
Std. Dev.	1.1809	0.7819	0.8972	1.1087	0.9418

Table 3: Statistical results of various indicators

Hook ID	Mean (μ)	Variance (σ^2)	Std. Dev. (σ)
1	0.0544	0.0977	0.3126
2	0.5088	0.2459	0.1568
3	1.4000	1.8872	1.3738
4	0.4576	0.8985	0.9479
5	0.5791	0.7027	0.8383
6	0.8290	1.2347	1.1112
7	0.1569	0.2956	0.5437
8	0.1451	0.3427	0.5854

Table 4: Number of psychological hooks for each sample

Statistic			Number
1 Hook			1339 (44.03%)
	2:626 (74.79%)		3:179 (21.39%)
Composite Hook	4:27 (3.22%)	5:4 (0.47%)	6:1 (0.1%)
			Total: 837
No Hook			865 (28.6%)

Table 5: Content source categories

Category	Count	Category	Count
keywords search	81 (2.66%)	recommend	364 (11.97%)
fashion	236 (7.76%)	food	308 (10.13%)
cosmetic	223 (7.33%)	movie and tv	259 (8.85%)
career	273 (8.98%)	love	307 (10.10%)
household	229 (7.53%)	gaming	253 (8.32%)
travel	269 (8.85%)	fitness	239 (7.86%)

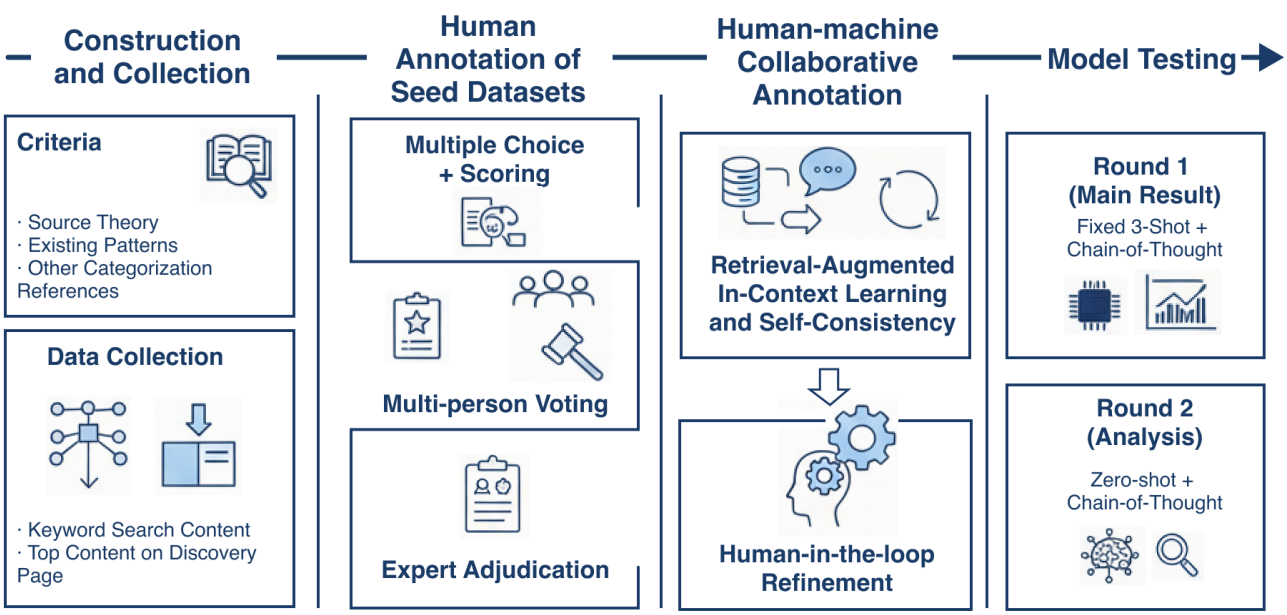

Figure 3: The Overall Construction Process

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Table 6: Dataset Statistics

(a) Overview

Statistic	Number
Total count	3041
Image count	1233 (40.55%)
Video count	1808 (59.45%)

(b) Title & Likes

Statistic	Number
Maximum title length	36
Average title length	15.05
Maximum like count	313,000
Average like count	7,248.79

Table 7: 8 psychological hooks counts & quality

Hook	Cnt	Hook	Cnt
Fear Of Missing Out (FOMO)	42	Perceptual Contrast	589
Gain Appeal	1153	Ingroup Identification / Outgroup Distinction	402
Information-gap	658	Social Comparison	102
Anomaly and novelty	226	Authority Endorsement	69
only high consensus			1620
high consensus and edge cases			341
only edge cases			215